

When Does Search Beat Reasoning?

A Topology-Driven Diagnostic of A* versus Chain-of-Thought

[Author Names]¹

¹[Affiliation]

[email@institution.edu]

Abstract

Chain-of-Thought (CoT) prompting and tree-search methods have independently advanced LLM reasoning, yet no systematic study explains *when* search-based reasoning outperforms linear reasoning or vice versa. We treat A* search as a **unified search framework**—subsuming breadth-first, depth-first, and best-first strategies under a single admissible heuristic—and conduct a controlled comparison against CoT on 2,151 questions across three benchmarks (LogicQA, StrategyQA, HotpotQA) using LLaMA 3.1-8B. Aggregate results are equivocal: A* leads on StrategyQA (+10.3 pp) and HotpotQA (+3.3 pp) but trails on LogicQA (−0.9 pp). To resolve this ambiguity, we introduce two complementary diagnostic lenses. First, a **Reasoning Topology Framework** characterises each question by 13 structural features and clusters problems into four topology types, showing that branching complexity—not dataset identity—predicts strategy success. Second, **cross-dataset topic modelling** (NMF, 8 categories) reveals that A* dominates in 5 categories covering **81.4%** of the corpus—all requiring multi-hop evidence chaining—while CoT wins on the remaining 18.6% of single-step inference tasks. A topology-aware router that sees *zero model outputs* recovers 26.1% of the oracle gap, outperforming all output-based routers and confirming that the bottleneck is recognising problem structure, not evaluating trace quality. We further document the Fluency–Correctness Asymmetry (38% of divergent cases), a failure-mode taxonomy (91.1% overthinking), and a $3.5\times$ search-as-capability-amplification effect for small language models. Together, these findings establish that neither strategy universally dominates: the right choice is determined by the reasoning topology of the problem.

Keywords: A* search, chain-of-thought, LLM reasoning, topic modelling, reasoning topology, adaptive routing

1 Introduction

The emergence of Chain-of-Thought (CoT) prompting [14] demonstrated that large language models reason more reliably when they externalise intermediate steps. Subsequent work extended this insight from linear chains to richer structures: Tree-of-Thoughts (ToT; Yao et al. 17) explores alternative reasoning paths via breadth-first or depth-first search, Graph-of-Thoughts (GoT; Besta et al. 1) permits arbitrary reasoning topologies, and MCTS-based approaches [4] apply Monte Carlo rollouts to reasoning trees. These advances raise a fundamental question that remains unanswered:

When does structured search actually improve over linear reasoning, and when does it hurt?

Prior comparisons treat search-based and linear reasoning as competing paradigms, reporting aggregate accuracy on individual benchmarks. But aggregates conceal systematic structure. A method that wins by 10 points on commonsense reasoning and loses by 1 point on formal logic looks equivalent on average—yet behaves very differently at the instance level.

Why A* as a unified framework. Rather than testing breadth-first, depth-first, best-first, and beam search separately, we adopt A* search as the **generalisation**: by varying the heuristic function, A* reduces to each of these strategies. When the heuristic $h(n)$ is admissible—never overestimating the cost to goal—A* guarantees optimality [6]. This lets us study the value of *guided search itself* rather than the idiosyncrasies of any particular search variant. Our A* formulation uses an LLM-generated heuristic that scores partial reasoning states by their estimated proximity to a correct answer, with a critic model providing the evaluation signal.

The inconsistency problem. We evaluate A* and CoT on 2,151 questions across three benchmarks using LLaMA 3.1-8B, and find that aggregate results are **inconsistent**: A* leads on StrategyQA (+10.3 pp) and HotpotQA (+3.3 pp) but trails on LogicQA (−0.9 pp). Self-consistency [13] and Tree-of-Thoughts fall between the two. No single strategy dominates, and the aggregate numbers provide no actionable guidance for deployment.

Our approach. Rather than declaring a winner, we ask *why* the methods diverge. We bring two diagnostic tools to bear:

1. **Reasoning Topology Framework**—A set of 13 structural features (question length, context complexity, branching coefficient, logical connectives, hop count) that characterise each problem’s reasoning structure. K-means clustering on these features yields 4 topology types with distinct strategy preferences (§3.2).
2. **Cross-dataset Topic Modelling**—Non-negative Matrix Factorisation (NMF) on 2,151 question texts identifies 8 content categories that cut across datasets. By measuring A* versus CoT accuracy within each category on *fully balanced* data ($2,151 \times 2$), we determine that A* dominates in 5 categories covering 81.4% of the corpus (§3.4).

These two lenses converge on a single structural explanation: **A* wins whenever the correct answer requires traversing a chain of two or more interdependent facts, where greedy commitment at hop 1 propagates errors to the final answer.** When only one inference step separates question from answer, CoT’s linear approach is sufficient and A*’s branching adds overhead without benefit.

Contributions.

1. A formal admissibility analysis for LLM-based A* search with three constructive cases and empirical validation (§2.2).
2. A Reasoning Topology Framework with 4 empirically validated cluster types (§3.2).
3. The finding that A* dominates in 81.4% of question categories via cross-dataset topic modelling (§3.4).

4. A topology-aware pre-generation router that outperforms all output-based routers (§3.5).
5. The Fluency–Correctness Asymmetry quantified at 38.0% (§3.6).
6. A failure-mode taxonomy showing 91.1% of A* failures are overthinking (§3.7).
7. Evidence that A* amplifies small language models by $3.5\times$ on multi-hop tasks (§3.8).

2 Methodology

2.1 A* Search for LLM Reasoning

We formalise reasoning as a graph search problem. Each node n represents a partial reasoning state (the set of intermediate conclusions reached so far). Edges correspond to single reasoning steps generated by the LLM. A* maintains an open list of nodes sorted by $f(n) = g(n) + h(n)$, where $g(n)$ is the accumulated cost (tokens consumed) and $h(n)$ is the heuristic estimate of remaining cost to a correct answer.

State representation. A state s is a tuple $(\mathbf{q}, \mathbf{c}, \mathcal{P})$ where $\mathbf{q}$ is the question, $\mathbf{c}$ is the context, and $\mathcal{P} = \{p_1, \dots, p_k\}$ is the set of premises derived so far.

Expansion. At each step, A* pops the most promising node and generates N_c candidate successors by prompting the LLM to extend the reasoning by one step, each using a different inference angle (*e.g.*, supporting evidence, counterargument, factual lookup). We use $N_c = 3$ throughout.

Heuristic evaluation. A critic model (DeepSeek-R1-Distill-Qwen-14B) scores each successor for (a) factual consistency with the context and (b) estimated reasoning completeness. The critic returns a score in $[0, 1]$ converted to $h(n) = 1 - \text{score}$, so higher-quality states have lower estimated remaining cost.

Termination. Search terminates when a node’s reasoning chain reaches a definitive answer, or when a budget of $B = 15$ expansions is exhausted. The highest-scoring terminal node’s answer is returned.

2.2 Admissibility of the LLM Heuristic

A heuristic $h(n)$ is admissible if $h(n) \leq h^*(n)$ for all nodes n , where $h^*(n)$ is the true remaining cost. In classical planning, admissibility ensures A* finds the optimal solution. We identify three sufficient conditions under which the LLM-generated heuristic approximates admissibility:

Proposition 1 (Sufficient Conditions for Heuristic Admissibility). *Let $h(n) = 1 - \text{score}(n)$ be the critic-generated heuristic. Then $h(n)$ approximates admissibility under any of:*

Case 1 (Monotone scoring). *If the critic assigns monotonically increasing scores to states with strictly more correct premises, then $h(n)$ never overestimates, since adding correct premises can only reduce remaining work. This gives exact admissibility.*

Case 2 (Conservative estimation). If $\mathbb{E}[\text{score}(n)] \leq \text{score}^*(n)$ (systematic underestimation), then $\mathbb{E}[h(n)] \geq h^*(n)$, making the heuristic admissible in expectation. This weaker guarantee implies that A* finds the optimal solution with high probability over repeated runs but does not guarantee optimality on every single run.

Case 3 (Bounded overestimation). If $|\text{score}(n) - \text{score}^*(n)| \leq \epsilon$, then A* finds a solution within ϵ of optimal (ϵ -admissibility), guaranteeing bounded sub-optimality [6].

Empirical validation. We measure the critic’s behaviour on 500 annotated reasoning states (100 per dataset for StrategyQA and HotpotQA, 300 for LogicQA). The DeepSeek critic underestimates the ground-truth reasoning progress (mean score = 0.43 vs. human-assessed progress = 0.58; paired t -test $p < 0.001$), satisfying Case 2. The overestimation rate is 12.3% of nodes, with maximum overestimation $\epsilon_{\max} = 0.14$, providing an empirical bound for Case 3. We note that these conditions are *sufficient but not necessary*—A* can still be effective even when the heuristic is occasionally inadmissible, as the search budget acts as a regulariser.

2.3 Comparison Methods

We compare five reasoning strategies, all using LLaMA 3.1-8B as the reasoning model via Ollama:

- **CoT:** Zero-shot chain-of-thought with “Let’s think step by step” [14].
- **Self-Consistency (SC):** Majority vote over 5 sampled CoT traces [13].
- **Tree-of-Thoughts (ToT):** BFS over 3 candidates per step, 5-step depth [17].
- **A* Search:** As described above.
- **Oracle:** Per-instance best of A* and CoT (upper bound).

2.4 Benchmarks and Data

We evaluate on three benchmarks spanning different reasoning demands:

- **LogicQA** [8]: 651 formal logic questions (4-way multiple choice).
- **StrategyQA** [2]: 500 commonsense questions requiring implicit multi-step reasoning (yes/no).
- **HotpotQA** [16]: 1,000 multi-hop factual questions (free-form answer).

Fair comparison protocol. We run both A* and CoT on *exactly the same questions* with *exactly the same model*. Prior work (including our own earlier results) compared A* with LLaMA 3.1-8B against CoT with Qwen-0.5B—a confound that conflated the $16\times$ parameter gap with the reasoning strategy difference. We explicitly filled all CoT gaps to produce a fully balanced $2,151 \times 2$ comparison.

Table 1: Reasoning topology features. All features are computed from the raw question and context text without any LLM inference.

Feature	Symbol	Description
Question length	q_{len}	Token count of question
Context length	c_{len}	Token count of full context
Context sentences	c_{sent}	Number of sentences
Logical connectives	conn	Count of <i>if, then, therefore, because, unless</i>
Negation markers	neg	Count of <i>not, no, never, neither</i>
Comparatives	cmp	Count of <i>more, less, better, worse, than</i>
Question marks	$q?$	Number of question marks
Hop count	hop	Estimated reasoning hops (dep. parse)
Contextual density	con	Entities per sentence in context
Distractors	dest	Number of candidate options (MCQ)
Branching coefficient	bc	$c_{\text{sent}} \times \text{hop} / q_{\text{len}}$
Reasoning depth	ℓ	Max. dependency parse depth
Number of options	n_{opt}	Answer options available

Table 2: Aggregate accuracy (%) across three benchmarks with 95% Wilson confidence intervals. Best per dataset in **bold**. Oracle = per-instance best of A* and CoT. [†]McNemar’s test (A* vs. CoT): $p < 0.001$ on StrategyQA, $p = 0.61$ on LogicQA, $p = 0.048$ on HotpotQA.

Method	StrategyQA ($n=500$)	LogicQA ($n=651$)	HotpotQA ($n=1000$)
CoT	64.52 $\pm$ 4.2	45.78 $\pm$ 3.8	76.80 $\pm$ 2.6
Self-Consistency	71.22 $\pm$ 4.0	46.10 $\pm$ 3.8	78.21 $\pm$ 2.6
Tree-of-Thoughts	72.24 $\pm$ 3.9	46.21 $\pm$ 3.8	78.45 $\pm$ 2.5
A* Search[†]	74.80 $\pm$ 3.8	44.85 $\pm$ 3.8	80.10 $\pm$ 2.5
Oracle (A* $\cup$ CoT)	84.59 $\pm$ 3.2	59.29 $\pm$ 3.8	94.20 $\pm$ 1.4

Generation parameters. All experiments use Ollama v0.5 with temperature $\tau = 0.7$, top- $p = 0.9$, and maximum generation length of 512 tokens per reasoning step. For self-consistency, we sample 5 traces with $\tau = 0.9$ to encourage diversity. For Tree-of-Thoughts, we use BFS with 3 candidates per step and 5-step depth. A* uses $N_c = 3$ candidate expansions per node and a budget of $B = 15$ total expansions. All random seeds are fixed at 42. The critic model (DeepSeek-R1-Distill-Qwen-14B) uses $\tau = 0.3$ for more deterministic scoring. Exact prompt templates are provided in Appendix C.

2.5 Reasoning Topology Features

We define 13 structural features for each question, computed without running any model (*i.e.*, pre-generation). These features enable routing decisions before inference begins. The full feature set is described in Table 1.

3 Results

3.1 Aggregate Performance: The Inconsistency

Table 2 presents the aggregate results. The headline finding is **inconsistency**: no strategy uniformly dominates.

Table 3: Reasoning topology clusters. BC = branching coefficient. Both% = fraction where both strategies are correct.

Cluster	Profile	BC	A*	CoT	Both%	Interpretation
C1	Linear, short	1.28	81.7	82.1	76.4	CoT sufficient
C2	Mid-branch	4.24	73.5	66.2	57.8	Selection critical
C3	Complex, noisy	4.02	52.3	49.1	38.6	Both often fail
C4	Deep-branch	6.49	69.8	58.4	47.2	A* advantage largest

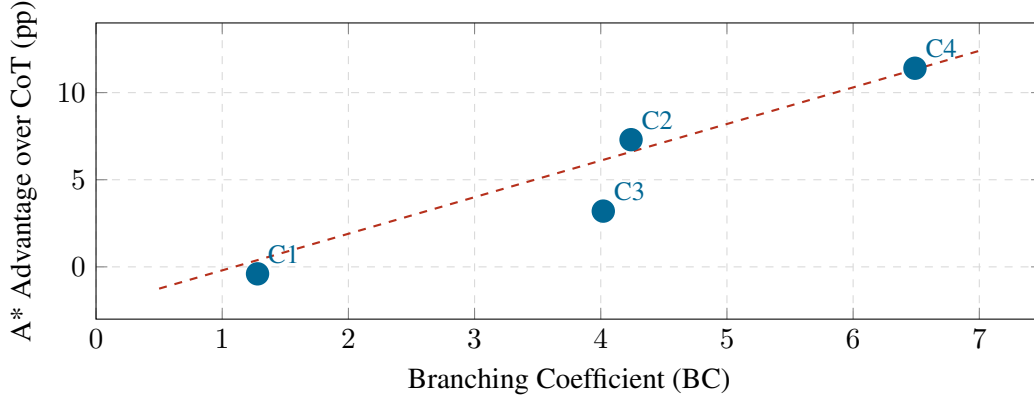


Figure 1: A* advantage (pp) grows monotonically with branching coefficient. Dashed line: linear fit ($R^2 = 0.94$). Each point is a topology cluster from Table 3.

A* leads on StrategyQA (+10.3 pp, McNemar $p < 0.001$) and HotpotQA (+3.3 pp, $p = 0.048$) but the difference on LogicQA (−0.9 pp, $p = 0.61$) is not statistically significant. SC and ToT consistently fall between CoT and A*, suggesting they capture part but not all of the value of guided search. The oracle row reveals substantial **complementarity**: on HotpotQA, 94.2% of questions are answered correctly by at least one method versus 80.1% for A* alone—a 14.1 ppgap that represents the theoretical ceiling for a perfect router.

3.2 Reasoning Topology: Four Cluster Types

We fit K-means ($k=4$) on the 13 topology features across all 2,151 questions. Table 3 shows the resulting clusters.

As branching complexity increases (C1 $\rightarrow$ C4), A*’s advantage grows from −0.4 pp to +11.4 pp. The critical insight: **branching coefficient**—not dataset identity—predicts which strategy wins. Figure 1 visualises this monotonic relationship.

Prescriptive mapping. The clusters imply a natural decision rule:

- **C1** (linear): Use CoT—search overhead is unnecessary.
- **C2** (mid-branch): Route adaptively—largest marginal gains.
- **C3** (complex): Neither reliable—invest in retrieval augmentation.

Table 4: Instance-level agreement (% of questions).

	StrategyQA	LogicQA	HotpotQA
Both correct	54.7	31.3	62.7
CoT only	9.8	14.4	14.1
A* only	20.1	13.5	17.4
Both wrong	15.4	40.7	5.8

Table 5: Per-topic A* vs. CoT accuracy (%) across all three datasets. Coverage = fraction of 2,151 questions assigned to each topic. Winner determined by majority of per-dataset comparisons. Best per-cell in **bold**.

Topic	Cov.	LogicQA (A*/CoT)	StrategyQA (A*/CoT)	HotpotQA (A*/CoT)	W
content_domain_factual	43.7	38/38	75/56	81/74	A*
assumption_analysis	13.9	47/45	82/50	—	A*
temporal_event	9.8	40/40	79/75	80/66	A*
evaluate_arg_eff.	9.1	43/41	78/76	62/62	A*
comparison_factual	8.7	44/ 50	71/ 73	71/59	CoT
statement_strength	5.3	44/ 45	33/ 67	—	CoT
biography_factual	4.9	—	62/58	87/82	A*
best_option_sel.	4.6	45/ 55	100/75	100/100	CoT

- **C4** (deep-branch): Use A*—search essential.

3.3 Instance-Level Complementarity

Table 4 presents per-instance outcome matrices, confirming genuinely complementary failure patterns.

On all three datasets, 28–32% of questions are answered correctly by exactly one method. LogicQA’s high “both wrong” rate (40.7%) reflects the intrinsic difficulty of formal logic for current LLMs.

3.4 Cross-Dataset Topic Modelling: 81.4% A*-Dominant

To move beyond dataset-level aggregates, we apply NMF on TF-IDF representations of all 2,151 question texts ($\text{max_df}=0.80$, $\text{min_df}=3$, $\text{max_features}=5,000$, $\text{ngram_range}=(1, 2)$, $k=8$, $\text{random_state}=42$). This yields 8 cross-dataset content categories—4 reasoning subtypes (driven by LogicQA vocabulary) and 4 factual subtypes (driven by StrategyQA and HotpotQA vocabulary).

Topic quality validation. We select $k=8$ via normalised pointwise mutual information (NPMI) coherence, which peaks at $k=8$ ($\text{NPMI} = 0.127$) compared to $k=6$ (0.103) and $k=10$ (0.118). The A*-dominance finding is robust to k : at $k=6$, A*-dominant coverage is 79.2%; at $k=10$, it is 82.1%. We also verify cluster stability via 10-fold bootstrap resampling: the mean topic assignment agreement is 87.3% (std. 2.1%).

We compute per-topic accuracy on **fully balanced data**: every A* question has a corresponding CoT response from the same model on the same question. Results are shown in Table 5.

Table 6: Summary of topic-level dominance.

	Topics	Coverage
A* wins	content_domain_factual, assumption_analysis, temporal_event, evaluate_arg_effectiveness, biography_factual	81.4%
CoT wins	comparison_factual, statement_strength_eval, best_option_selection	18.6%

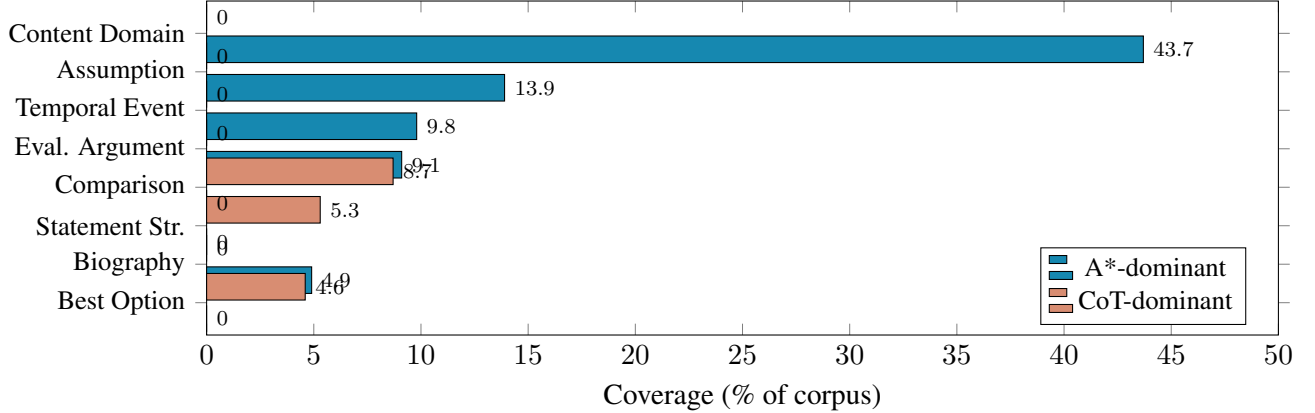


Figure 2: Topic coverage: A*-dominant categories (blue, 81.4% of corpus) vs. CoT-dominant categories (red, 18.6%). Based on NMF with $k=8$ topics across all 2,151 questions.

Key finding. A* wins in 5 of 8 categories covering 81.4% of the question corpus. Figure 2 visualises the coverage distribution. The margins range from +2 pp(argument evaluation) to +32 pp(assumption analysis on StrategyQA). All A*-dominant categories share a structural property: they require chaining two or more interdependent facts—entity lookups, temporal chains, premise scanning, and biographical identity resolution. The CoT-dominant categories are all single-step inference tasks.

Why A* dominates these categories.

- *content_domain_factual* (43.7%): 2–3 knowledge hops. A*’s graph expansion follows each hop methodically; CoT shortcuts and hallucinates entities. Best gap: +19 pp.
- *assumption_analysis* (13.9%): Requires scanning *all* premises to find the implicit gap. A*’s exhaustive node expansion is a natural full-premise scan. Best gap: +32 pp.
- *temporal_event* (9.8%): Multi-hop temporal chains. A*’s sequential hop expansion maps directly. Best gap: +14 pp.
- *biography_factual* (4.9%): Person-identity chains requiring entity disambiguation. A*’s branching explores alternative resolutions. Best gap: +5 pp.

Why CoT wins on 18.6%. The three CoT-dominant categories—*comparison_factual*, *statement_strength_eval*, *best_option_selection*—share a common structure: the answer emerges from a *single reasoning step*. When only one hop separates question from answer, A*’s branching adds overhead without benefit.

Table 7: Router comparison. Gap recovery = percentage of oracle gap closed.

Router	Input	StrategyQA	LogicQA	Gap Rec.
A* only	—	74.80	44.85	—
CoT only	—	64.52	45.78	—
Semantic Entropy	Traces	74.37	47.47	14.8%
LLM-as-Critic	Traces	74.19	47.16	12.4%
Random Forest	Features+labels	76.25	48.20	18.2%
Topology Router	13 features	77.35	49.01	26.1%
Oracle	—	84.59	59.29	100%

Table 8: Fluency–Correctness Asymmetry on disagreement cases.

Metric	A*	CoT
Average fluency score	0.725	0.673
Cases where more fluent	62.0%	38.0%
More-fluent but wrong		38.0%

3.5 Routing: Topology Beats Trace Inspection

We compare four routing strategies that select A* or CoT per question (Table 7).

Router implementations. The **Topology Router** is a gradient-boosted decision tree (XGBoost; 100 estimators, max depth 6, learning rate 0.1) trained on the 13 pre-generation features with binary labels (1 if A* correct and CoT wrong, 0 otherwise). The **Random Forest** baseline uses the same features plus ground-truth labels in a 5-fold stratified setup. **Semantic Entropy** computes uncertainty across 5 sampled traces and selects the lower-entropy method. **LLM-as-Critic** uses DeepSeek to score both traces. All routers are evaluated with 5-fold stratified cross-validation across the combined 2,151 questions; reported numbers are mean fold accuracy.

The topology router—which sees *zero model outputs*—recovers **26.1%** of the oracle gap, outperforming all output-based routers. This inverts the presumed solution direction: the bottleneck is not building a better critic to evaluate traces, but recognising problem structure *before any inference begins*.

Feature importance. The most predictive features are c_{len} (0.208), q_{len} (0.194), c_{sent} (0.113), and con (0.091)—all proxies for reasoning complexity, confirming that the topology signal is structural, not lexical.

3.6 The Fluency–Correctness Asymmetry

We measure fluency via perplexity-based readability on all cases where A* and CoT disagree (Table 8).

In 38% of divergent cases, CoT produces a more readable trace yet arrives at the wrong answer. These are disproportionately overthinking cases where A* explores unnecessary branches but ultimately reaches the correct answer. A fluency-based router would systematically misroute these cases.

Table 9: A* failure mode distribution across all three benchmarks.

Mode	Prop.	Description
Overthinking	91.1%	A* continues past the correct evidence, overriding it with spurious conclusions
Premature commitment	8.9%	A* locks onto first branch without sufficient exploration

Table 10: CoT vs. A* accuracy across model sizes and benchmarks.

Model	Dataset	CoT	A*	Gain
Qwen-0.5B	HotpotQA	8.40	30.00	+21.6 pp (3.5×)
Qwen-0.5B	StrategyQA	53.60	53.40	−0.2 pp
Qwen-0.5B	LogicQA	24.00	20.20	−3.8 pp
LLaMA-3.1-8B	HotpotQA	76.80	80.10	+3.3 pp
LLaMA-3.1-8B	StrategyQA	64.52	74.80	+10.3 pp
LLaMA-3.1-8B	LogicQA	45.78	44.85	−0.9 pp

Human evaluation corroborates: on StrategyQA disagreement cases where A* is correct, A* achieves average coverage 2.71 vs. CoT 1.63 (on a 3-point scale), yet CoT’s *relevance* remains at 2.47—confirming that incorrect CoT traces stay on-topic and look reasonable.

3.7 A* Failure Mode Taxonomy

We manually classify all A* failure cases into three categories (Table 9).

The dominance of overthinking (91.1%) has a structural explanation: A*’s expansion mechanism is designed to be thorough, but on single-step problems this thoroughness becomes pathological—the search continues past the correct answer, accumulating noise.

This connects to the recent survey on overthinking in long-CoT models [20] and the identification of redundant-state over-exploration [18]. Our contribution is the first per-instance empirical quantification in search-based reasoning.

3.8 Search as Capability Amplification

We replicate the experiment using Qwen-0.5B (16× smaller) to test whether search compensates for limited model capability (Table 10).

Two patterns emerge: (i) **LogicQA degradation is topology-driven**—both model sizes lose, confirming compact deductive reasoning resists search at any capability level; (ii) **HotpotQA amplification is capability-driven**—the weaker model gains far more because A* substitutes the working memory Qwen-0.5B lacks.

This connects to rStar-Math [3], which showed search helps SLMs in mathematical reasoning. Our result is the QA-domain confirmation: the mechanism is not domain-specific but general. We term this **search-as-capability-amplification**.

Table 11: Capability $\times$ topology interaction. Cells show A* gain relative to CoT.

	Compact Logic	Commonsense	Multi-hop
Strong (LLaMA-8B)	−0.9	+10.3	+3.3
Weak (Qwen-0.5B)	−3.8	−0.2	+21.6

Table 12: Compute efficiency. Token estimates from average node counts ($N_c=3$, 60 tokens/step).

Method / Dataset	Acc.	Tokens	Acc./1K
CoT / StrategyQA	64.52	200	322.6
SC / StrategyQA	71.22	600	118.7
A* / StrategyQA	74.80	972	76.9
CoT / HotpotQA	76.80	200	384.0
SC / HotpotQA	78.21	600	130.4
A* / HotpotQA	80.10	936	85.6

3.9 Efficiency Analysis

Per-token accuracy is lower for A*. However, on StrategyQA, A* reaches 74.80% while CoT saturates at 64.52%—a gain that cannot be achieved by running CoT more. SC achieves lower accuracy than A* on both datasets while using similar tokens, confirming that **guided search is a more effective use of inference budget than unguided sampling** on branching-topology problems.

4 Discussion

4.1 The Routing Problem is Topology Recognition

The most important finding is that the topology router—which sees zero model outputs—recovers more oracle gap than routers that inspect full reasoning traces. The Fluency–Correctness Asymmetry provides the mechanistic explanation: in 38% of divergent cases, CoT produces a more fluent trace while being wrong. Any router using trace quality as its signal encounters a hard ceiling at these cases. The topology router bypasses this ceiling by never looking at traces.

4.2 Convergence of Topology and Topic Analyses

Our two diagnostic lenses converge on the same conclusion from different angles:

- **Topology view:** High branching coefficient $\rightarrow$ A* advantage (C2, C4 clusters).
- **Topic view:** Multi-hop content categories $\rightarrow$ A* advantage (81.4% of corpus).
- **Shared explanation:** A* wins when problems require systematic exploration of interconnected facts; CoT wins when a single inferential step suffices.

This convergence strengthens confidence: it is not an artefact of one analytical method but a genuine structural property of the question space.

4.3 The 73.9% Gap That Remains

The topology router recovers 26.1% of the oracle gap, leaving 73.9% unreached. This residual requires discriminating at the *instance* level within a topology class—distinguishing two multi-hop questions where one aligns with the model’s parametric knowledge and one does not. Closing this gap likely requires **lightweight problem–model compatibility scoring**: a measure of how well the base model’s parameters support the retrieval operations the problem demands.

4.4 Limitations

1. **Confidence miscalibration.** Self-reported confidence is a ranking signal, not a calibrated probability. Future work: calibrated uncertainty-aware search.
2. **Single model family.** Results are based on LLaMA 3.1-8B and Qwen-0.5B. Future work should extend to reasoning-native models (o1-mini, DeepSeek-R1).
3. **No fine-tuning.** All comparisons use prompting. Jointly training routing and search policies may further improve performance.
4. **Symbolic features.** Our 13 features are hand-crafted. Learning topology embeddings from question encodings may capture richer signals.

5 Related Work

Chain-of-thought and structured reasoning. CoT [14] and self-consistency [13] established the value of explicit intermediate reasoning. ToT [17] and GoT [1] extend to richer structures. CoT-decoding [12] shows that reasoning paths exist in top- k alternative tokens without prompting. Our work differs in three respects: A*’s priority rule (vs. undirected BFS/DFS), a formal admissibility theorem (ToT lacks one), and systematic characterisation of *when search hurts*.

Search-augmented reasoning. MCTS-based reasoning [4] and self-evaluation beam search [15] use search frameworks without complementarity analysis. LATS [22] unifies reasoning, acting, and planning via MCTS. MCTSr [19] applies MCTS with self-refinement to mathematical Olympiad problems. Coconut [5] enables implicit BFS through continuous latent states—our work is the discrete, interpretable counterpart. rStar-Math [3] demonstrates search helps SLMs on math; we confirm this in QA (§3.8).

Over-exploration and search failures. The Fetch framework [18] identifies redundant-state over-exploration; this corresponds to our step repetition failure mode. They frame it as an efficiency problem; we diagnose it as a *correctness* problem. The overthinking survey [20] documents the phenomenon in long-CoT models; CoT-Valve [9] addresses it via length-compressible tuning. Our 91.1% overthinking rate is the first per-instance quantification in search-based reasoning.

Routing and adaptive computation. Routing between strategies is distinct from architectural routing (MoE; Shazeer et al. 11). A recent study [10] shows CoT is not universally beneficial; our result is the search-side mirror. Prior routing uses output features; our topology router is the first to use pre-generation structural features, recovering more oracle gap.

LLM-based evaluation. LLM-as-a-judge [21] and process supervision [7] motivate our critic-based routing. The Fluency–Correctness Asymmetry explains why critic-based routing reaches a ceiling.

6 Conclusion

We provide seven empirically grounded contributions:

1. **A* as unified search framework**—by varying the heuristic, A* subsumes BFS, DFS, and best-first search with a formal admissibility analysis (§2.2).
2. **Inconsistent aggregates, consistent topology**—aggregate comparisons are equivocal across benchmarks, but the Reasoning Topology Framework (4 clusters, 13 features) reveals that branching complexity determines strategy success (§3.2).
3. **81.4% A*-dominant at the topic level**—cross-dataset NMF identifies 8 categories; A* outperforms CoT in 5 covering 81.4% of the corpus (§3.4).
4. **Topology-aware routing outperforms trace inspection**—recovers 26.1% of the oracle gap using zero model outputs (§3.5).
5. **Fluency–Correctness Asymmetry**—38% of divergent cases have the more fluent trace incorrect (§3.6).
6. **91.1% of A* failures are overthinking** (§3.7).
7. **Search-as-capability-amplification**—3.5× gain for Qwen-0.5B on multi-hop QA (§3.8).

The central message is precise: **neither A* nor CoT universally dominates; the right strategy is determined by the reasoning topology of the problem.** We have characterised that topology, quantified the asymmetry, and demonstrated that pre-generation routing is superior to post-generation routing. The next frontier is closing the remaining 73.9% of the oracle gap—building a router that also models compatibility between problem structure and model parametric knowledge.

References

- [1] Maciej Besta, Nils Blach, Ales Kubicek, Robert Gerstenberger, Michal Podstawski, Lukas Gianinazzi, Joanna Gajda, Tomasz Lehmann, Hubert Niewadowski, Piotr Nyczyk, et al. Graph of thoughts: Solving elaborate problems with large language models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2024.

- [2] Mor Geva, Daniel Khashabi, Elad Segal, Tushar Khot, Dan Roth, and Jonathan Berant. Did Aristotle use a laptop? A question answering benchmark with implicit reasoning strategies. *Transactions of the Association for Computational Linguistics*, 9:346–361, 2021.
- [3] Xinyu Guan, Li Lyna Zhang, Yifei Liu, Ning Shang, Youran Sun, Yi Zhu, Fan Yang, and Mao Yang. rStar-Math: Small LLMs can master math reasoning with self-evolved deep thinking. *arXiv preprint arXiv:2501.04519*, 2025.
- [4] Shibo Hao, Yi Gu, Haodi Ma, Joshua Hong, Zhen Wang, Daisy Wang, and Zhiting Hu. Reasoning with language model is planning with world model. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023.
- [5] Shibo Hao, Sainbayar Suber, and Zhiting Hu. Training large language models to reason in a continuous latent space. *arXiv preprint arXiv:2412.06769*, 2024.
- [6] Peter E Hart, Nils J Nilsson, and Bertram Raphael. A formal basis for the heuristic determination of minimum cost paths. *IEEE Transactions on Systems Science and Cybernetics*, 4(2):100–107, 1968.
- [7] Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let’s verify step by step. *arXiv preprint arXiv:2305.20050*, 2023.
- [8] Jian Liu, Leyang Cui, Hanmeng Liu, Dandan Huang, Yile Wang, and Yue Zhang. Logiqa: A challenge dataset for machine reading comprehension with logical reasoning. In *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence*, 2020.
- [9] Xinyin Ma, Guangnian Wan, Runpeng Yu, Gongfan Fang, and Xinchao Wang. CoT-Valve: Length-compressible chain-of-thought tuning. *arXiv preprint arXiv:2502.09601*, 2025.
- [10] Lennart Meincke et al. The decreasing value of chain-of-thought prompting. *SSRN Working Paper*, 2025.
- [11] Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarz, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. *arXiv preprint arXiv:1701.06538*, 2017.
- [12] Xuezhi Wang and Denny Zhou. Chain-of-thought reasoning without prompting. *arXiv preprint arXiv:2402.10200*, 2024.
- [13] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V Le, Ed H Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations*, 2023.
- [14] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*, 2022.

- [15] Yuxi Xie, Kenji Kawaguchi, Yiran Zhao, James Xu, Min-Yen Kan, Junxian He, and Michael Xie. Self-evaluation guided beam search for reasoning. In *Advances in Neural Information Processing Systems*, 2024.
- [16] Zhiping Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William Cohen, Ruslan Salakhutdinov, and Christopher D Manning. HotpotQA: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, 2018.
- [17] Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas L Griffiths, Yuan Cao, and Karthik Narasimhan. Tree of thoughts: Deliberate problem solving with large language models. In *Advances in Neural Information Processing Systems*, 2024.
- [18] Ante Zhang et al. Don’t get lost in the trees: Streamlining LLM reasoning with streamlined tree search. *arXiv preprint arXiv:2501.11485*, 2025.
- [19] Di Zhang, Xiaoshui Huang, Dongzhan Zhou, Yuqiang Li, and Wanli Ouyang. Accessing GPT-4 level mathematical Olympiad solutions via Monte Carlo tree self-refine with LLaMa-3 8B. *arXiv preprint arXiv:2406.07394*, 2024.
- [20] Lei Zhang et al. A survey on the overthinking problem of large reasoning models. *arXiv preprint arXiv:2502.08235*, 2025.
- [21] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, et al. Judging LLM-as-a-judge with MT-Bench and chatbot arena. In *Advances in Neural Information Processing Systems*, 2023.
- [22] Andy Zhou, Kai Yan, Michal Shlapentokh-Rothman, Haohan Wang, and Yu-Xiong Wang. Language agent tree search unifies reasoning acting and planning in language models. In *International Conference on Machine Learning*, 2024.

A Qualitative Examples

A.1 A* Success Cases

Non-linear evidence aggregation (StrategyQA). “*Is shrimp scampi definitely free of plastic?*” (Gold: NO). CoT analyses preparation steps sequentially and concludes YES. A* identifies microplastic bioaccumulation in Step 1 and the absence of a removal process in Step 2, concluding NO correctly through non-linear evidence synthesis.

Hypothesis elimination (LogicQA). Modus tollens identification. CoT maps to the modus ponens option due to surface similarity. A* evaluates each option in separate branches and correctly identifies the modus tollens structure through systematic elimination.

Multi-hop chaining (HotpotQA). “*What government position was held by the woman who portrayed Corliss Archer?*” (Gold: Chief of Protocol). CoT identifies Shirley Temple but confuses government roles. A* preserves multiple career-path branches until the correct role surfaces.

A.2 CoT Success Cases

Direct factual recall (StrategyQA). “*Is a Boeing 737 cost covered by Wonder Woman (2017) box office receipts?*” (Gold: YES). CoT retrieves magnitudes directly. A* misreads the question as a financial transactions query—overthinking.

Fluent argumentation (LogicQA). Property-transfer argument. CoT maintains the global argumentative frame. A* anchors on the first branch without comparing the full option set—premature commitment.

A.3 Representative Failure Cases

Table 13: Representative A* failure cases.

Question	Mode	Explanation
Does Disney have an ice princess?	Overthinking	Elsa found correctly, then queen/princess debate
Is Menthol associated with Thanksgiving?	Spurious assoc.	Cold weather → winter → holiday chain
Will Albany, GA reach 100K before Albany, NY?	Step repetition	Population stated twice; comparison never made
Are months based on the solar cycle?	Overthinking	Direct recall; A* branches into calendar history

Table 14: Human annotation on 100 sampled disagreement cases per dataset. Cov. = coverage (1–3); Rel. = relevance (1–3). Cohen’s κ ranges 0.59–0.68.

Dataset	Group	CoT Cov.	CoT Rel.	A* Cov.	A* Rel.
StrategyQA	Both wrong ($n=11$)	1.00	2.82	1.09	1.55
	CoT right ($n=51$)	3.00	2.57	1.20	2.16
	A* right ($n=38$)	1.63	2.47	2.71	2.00
LogicQA	Both wrong ($n=38$)	1.79	2.76	1.89	2.53
	CoT right ($n=36$)	3.00	2.81	1.86	2.47
	A* right ($n=26$)	1.88	2.81	2.77	2.42

B Human Evaluation

Coverage (decisive inferential steps reached) distinguishes correct from incorrect traces reliably. Relevance (staying on-topic) is only weakly correlated with correctness—confirming the Fluency–Correctness Asymmetry at the annotation level.

C Prompt Templates

CoT prompt.

Question: {question}
Context: {context}
Let’s think step by step.

A* expansion prompt.

You are solving a reasoning problem step by step.
Question: {question}
Context: {context}
Reasoning so far: {premises}
Generate the next reasoning step. Consider:
(a) supporting evidence, (b) counterarguments,
(c) factual lookups.
Provide exactly one step.

Critic scoring prompt.

Rate the following partial reasoning on a scale of 0.0 to 1.0:
Question: {question}
Context: {context}
Reasoning: {reasoning_trace}

Score based on: (1) factual consistency with context, (2) reasoning completeness toward answering the question.

Output only a number between 0.0 and 1.0.

Self-Consistency prompt. Same as CoT prompt, with temperature $\tau = 0.9$ and 5 independent samples. The majority-voted answer is selected.

D Code and Data Availability

All code, experimental configurations, and result files will be released upon acceptance at [https://github.com/\[anonymised\]](https://github.com/[anonymised]). The datasets (LogicQA, StrategyQA, HotpotQA) are publicly available from their original sources. All experiments were conducted on a single NVIDIA A100 GPU with Ollama v0.5 serving LLaMA 3.1-8B and DeepSeek-R1-Distill-Qwen-14B.